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Optic flow and cycling effort: Where to look to go faster

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ABSTRACT

Optic flow can significantly influence the perception and exertion of effort. In this study, we investigated the effects of exposure to proximal and distal areas of the optic flow field on exerted and perceived cycling effort. Thirty cyclists participated in two 20-min cycling trials within a virtual reality environment, with the goal of surpassing the power output achieved during a baseline trial. During these trials, they viewed the environment through a proximal or distal window, in counterbalanced order. We measured the cyclists' exerted effort on a bicycle trainer, and recorded their responses regarding perceived effort and psychological momentum experience. A one-way repeated measures ANCOVA with average baseline power as a covariate revealed a significant difference in exerted effort between the proximal and distal condition, with higher average exerted effort in the proximal condition. However, a significant interaction effect between condition and baseline power indicated that the beneficial effect of the proximal condition was mainly present for lower-level cyclists. We observed no significant differences in perceived effort or psychological momentum. These findings provide novel insights into the relation between optic flow and cycling effort, and call for new research on the mechanisms underlying this relation.

1. Introduction

During cycling trials, athletes constantly need to evaluate whether to maintain, increase, or decrease their efforts (Foster et al., 2004; Smits et al., 2014). The process of regulating these efforts does not exclusively result from internal bodily signals, such as muscle fatigue and heart rate, but is also informed visually (Smits et al., 2014; Tucker & Noakes, 2009). This information can be presented to athletes in terms of numbers, such as current speed, cadence, or power output (Smits et al., 2016), but is also contained in the optic flow related to the person's movement through the environment (Parry et al., 2012). In other words, the structural visual motion of elements in the environment resulting from observers' movements informs them about their speed. Manipulating optic flow velocity has demonstrated to influence perceived and actual exertion of cycling effort (Parry et al., 2012).

The optic flow of a moving observer is characterised by fast-flowing proximal elements and slower-flowing distal elements (Gibson et al., 1959). This implies that while cycling, gazing nearby on the road exposes cyclists to higher flow velocities than gazing far away. While previous research has primarily focused on manipulating the velocity of the complete flow field (Bardy et al., 1992; Baumberger

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et al., 2000; Guerin & Bardy, 2008; Mohler et al., 2007; Pailhous et al., 1990; Parry et al., 2012), no study has yet investigated effects of exposure to specific parts of the flow field, such as the proximal and distal areas. Furthermore, no study has explored these effects within sports performance contexts. Here, we aimed to take the first step by examining these effects within the context of individual cycling trials. In the next section we will first outline the current knowledge on the relationship between optic flow velocity and effort regulation, and will then explore how exposure to proximal elements might lead to a different sense of egospeed (i.e., the speed of self-motion; Eggleston et al., 1999). Subsequently, we will discuss how exposure to proximal elements in the optic flow might lead to a sense of positive psychological momentum (PM), which could ultimately result in increased effort during cycling trials. This idea will then be tested empirically.

1.1. The role of optic flow in locomotion and effort regulation

During locomotion, optic flow serves a proprioceptive role by providing information about the observer's direction and speed (Gibson, 1966). This information, in turn, influences displacement through a continuous dynamic coupling between perception and action (Fajen, 2021; Gibson et al., 1955; Warren, 2006). An example is the coupling between optic flow velocity and locomotion speed. Several studies have demonstrated that experimentally speeding up optic flow leads walking individuals to reduce energy expenditure (Guerin & Bardy, 2008) and walking speed (Bardy et al., 1992; Baumberger et al., 2000; Mohler et al., 2007; Pailhous et al., 1990). In brief, when optic flow specifies an egospeed greater than the individual's physical locomotion speed, individuals who are instructed to walk at a natural pace or at a pace of around 5 to 9 km/h tend to slow down.

Similarly, Parry et al. (2012) demonstrated that the velocity of optic flow influences the perception and regulation of cycling effort. In their study, cyclists were asked to perform three time trials on a bicycle trainer, in which they were instructed to match their cycling behaviour to the average power output and cadence from a baseline trial. During the three test trials, the cyclists were exposed to a virtual environment that unfolded at a velocity that either matched their actual speed, was reduced in speed by 15 %, or was sped up by 15 %. When the cyclists experienced reduced optic flow velocity, they reported lower ratings of perceived exertion (RPE; Borg et al., 1987) throughout the trial compared to when the optic flow velocity was increased or matched their actual speed. Moreover, the cyclists in this study also exerted significantly more effort during the trial with reduced optic flow velocity. Together with the aforementioned studies on walking (Bardy et al., 1992; Baumberger et al., 2000; Guerin & Bardy, 2008; Mohler et al., 2007; Pailhous et al., 1990) these findings demonstrate a consistent negative association between the direction of optic flow velocity manipulations and locomotion speed, when the instruction was to walk or cycle at a preferred or previously defined speed. Accordingly, Parry et al.'s (2012) findings suggest that optic flow may play a role in a larger system regulating effort exertion, alongside other factors such as body temperature, afferent feedback from cardiovascular, respiratory, and musculoskeletal systems, and psychological factors such as emotions and prior experiences (Bruce et al., 2019; Marcora, 2009; Tucker, 2009; Williamson, 2010).

While research on the effects of optic flow velocity on locomotion speed and effort exertion has primarily focused on manipulating the velocity of the complete flow field (e.g., Bardy et al., 1992; Baumberger et al., 2000; Guerin & Bardy, 2008; Mohler et al., 2007; Pailhous et al., 1990; Parry et al., 2012), locomotion speed and effort exertion might also depend on which areas of the optic flow are available to the observer. Due to the faster flow of proximal elements in the optic array compared to distal elements (Gibson et al., 1959), exposure to a proximal area of optic flow exposes observers to faster flowing elements than exposure to a distal area. Given that the perception of egospeed relies on the flow velocity present in the observer's visual field (Lidestam et al., 2019), this raises the question of whether optic flow exposure could evoke effects on locomotion speed and effort exertion.

In one of the rare studies on this topic, the effect of proximal and distal exposure on egospeed perception was implied when participants estimated which virtual optic flow velocity matched their walking speed (Banton et al., 2005). This task involved two conditions: one where participants maintained a straight-ahead gaze and another where they directed their gaze downward. When participants gazed straight ahead, they selected a virtual optic flow that was faster (and less accurate) than when gazing nearby, suggesting a perceived need to increase optic flow velocity to match their walking speed. In other words, the participants seemed to feel that the straight-ahead gaze indicated an egospeed that was too slow to match their physical walking speed. Although the authors did not take into account differences in optic flow velocity, and did not examine effects on walking speed or energy expenditure, results did suggest that exposure to proximal areas might lead observers to perceive their speed as higher compared to exposure to distal areas.

1.2. Optic flow velocity indicating goal progression and psychological momentum

In the sports context, which is the focus of this study, no research on optic flow and effort has been conducted thus far. Parry et al. (2012) instructed their cyclists to match their effort and cadence to that of a baseline trial. Based on their results, the authors concluded that reduced optic flow velocity could imply an effort level that is too low to match the baseline trial, necessitating increased effort. However, researchers have indicated that studying the effects of visual perception on effort regulation should be conducted within a realistic context (Araujo et al., 2006; Smits et al., 2014). Indeed, Smits et al. (2014) emphasised that the relationship between environmental information and effort regulation should be considered in light of task demands and the athlete's goals. In sports such as cycling, where performance is assessed in terms of locomotion speed, the goal often involves outperforming oneself or others. In the present study, we will create such a context by examining cyclists in situations in which they need to outperform themselves. Thus, by creating this context, we deviate from previous optic flow research (e.g., Bardy et al., 1992; Baumberger et al., 2000; Guerin & Bardy, 2008; Mohler et al., 2007; Pailhous et al., 1990; Parry et al., 2012), where the instruction was to walk or cycle at a preferred or previously defined speed, which could potentially result in a different optic flow – effort relationship (Smits et al., 2014).

Because optic flow velocity specifies egospeed (Dyre, 1997; Larish & Flach, 1990; Warren, 1982), it could arguably inform athletes about their goal progression in sports where performance is based on egospeed, such as cycling or running. Such perceptions of progress (or regress) in relation to a goal give rise to positive (or negative) changes in cognition, affect, motivation, physiology, and behaviour, known as PM (Adler, 1981; Gernigon et al., 2010; Markman & Guenther, 2007; Taylor & Demick, 1994; Vallerand et al., 1988). Especially in sports contexts in which people exert significant effort, they are sensitive to changes in PM (Perreault et al., 1998). For instance, Den Hartigh et al. (2014) found that rowers competing against virtual opponents could more easily maintain their effort when manipulations led them to believe that they were gradually catching up (positive PM) than when they believed that they were gradually falling behind (negative PM). In the latter case, their efforts dropped more sharply. Moreover, research shows that cyclists can exert more effort during positive PM scenarios compared to neutral scenarios (Perreault et al., 1998). While these examples involved the presence of opponents to provide information about goal progression, performing in the absence of an opponent can also generate positive or negative PM depending on the goal progression one experiences (Markman & Guenther, 2007).

Given this background, exposure to a proximal visual field comes with exposure to faster flow which could lead individuals to perceive that they are moving faster, thereby making them believe they are progressing faster toward their goal, and thus triggering positive PM (Adler, 1981; Gernigon et al., 2010; Markman & Guenther, 2007; Taylor & Demick, 1994; Vallerand et al., 1988). With this in mind, we expect that the perception of faster optic flow in a proximal scene results in heightened levels of effort exertion for cyclists in the pursuit of their goal (cf. Briki et al., 2012; Den Hartigh et al., 2014).

1.3. The present study

The present study is a first step in examining the effects of exposure to proximal and distal elements in optic flow on exerted and perceived effort in cycling trials. In doing so, we also examined the possible role of PM in the relationship between proximal and distal optic flow exposure and effort. Within a virtual reality (VR) environment, participants were asked to outperform the average power output of a previous VR trial, while being exposed to either the proximal or distal elements in their visual field. In this way, we created a clear, challenging goal, which provides the appropriate conditions for PM to emerge (Adler, 1981; Gernigon et al., 2010; Markman & Guenther, 2007; Taylor & Demick, 1994; Vallerand et al., 1988).

In line with the methods used by Mohler et al. (2007) and Parry et al. (2012), the visual motion in the virtual scene was dynamically coupled to the cyclists' power output, allowing for a realistic virtual experience. However, instead of using projected video footage, we used a VR head-mounted display to fully immerse the participants in the scene. Throughout the trials, we measured power output as an indicator of exerted effort, and participants provided verbal ratings of their RPE (Parry et al., 2012) and PM experience (Briki et al., 2013; Den Hartigh et al., 2016; Perreault et al., 1998). Our hypothesis was that cyclists would demonstrate higher exerted and perceived effort in the proximal condition compared to the distal condition. Secondly, we hypothesized that this relationship would be influenced by the cyclists' PM experience. Specifically, we expected that participants who were exposed to the proximal elements in their visual field would experience more positive PM and that positive PM would be related to higher levels of exerted effort.

2. Methods

2.1. Participants

A G*Power analysis for a repeated measures ANOVA with two measurements (proximal vs. distal) indicated that a minimum of 34 participants was needed to achieve a statistical power of 0.8, assuming a medium effect size ($f = 0.25$) and an alpha level of 0.05. We recruited 37 cyclists between 18 and 60 years old, with normal or corrected-to-normal vision, who were familiar with performing cycling trials on a regular basis. Participants were recruited through online platforms, cycling clubs, and university faculties. Of the 37 participants, six were unable to complete all trials. Additionally, one participant who completed all trials was excluded from the analysis after suffering an injury between the distal and proximal trials, which resulted in a notable decrease in exerted effort after recovery. The remaining sample of 30 cyclists (83.3 % male), had a mean age of 33.1 years ($SD = 12.5$), a mean height of 174.1 cm ($SD = 8.9$), and a mean weight of 67.9 kg ($SD = 9.6$). The participants had experience in road cycling ($n = 19$), triathlon ($n = 7$), or mountain biking ($n = 4$) for a minimum of 1 year ($M = 8.45$, $SD = 6.93$), and trained at least 1 day per week ($M = 3.92$, $SD = 4.41$), accumulating a minimum training time of 1 h per week ($M = 5.65$, $SD = 5.71$). Participation in the study was voluntary, without compensation, and all participants provided informed consent prior to their involvement. The study received ethical approval from the Institutional Review Board of EuroMov Digital Health in Motion (IRB-EM 2201 J).

2.2. Materials

The participants performed a baseline trial and two "windowed" (proximal and distal) trials on the same bicycle (a B'TWIN or their own). Prior to the first trial, we adjusted the saddle height for participants using the B'TWIN bicycle by allowing them to try the bicycle and adjusting it until it felt right. We measured the chosen saddle height to ensure consistency across all trials. For participants using their own bicycles, we asked them to maintain the same saddle height setting throughout the study. The bicycles were mounted on a Tacx NEO T2 Smart bicycle trainer (Wassenaar, The Netherlands), which measured instantaneous power output in Watts at a frequency of approximately 2.4 Hz, with a deviation of less than 1 %. The bicycle trainer's resistance was set to simulate a horizontal ride, with the slope set at 0 which we kept consistent across all trials. Participants had the option to manually adjust the gears to their preference during the trials.

The participants wore an HTC Vive Pro head-mounted display that presented a virtual environment created using Unity

2019.1.10f1 with C# as programming language. The Unity project ran on a computer with Windows 10, a 64-bit operating system, an Intel Core i7 -7700HQ, 2.8 GHz, 16 GB RAM, and a NVIDIA GeForce GTX 1050 graphics card. The virtual environment consisted of a straight and flat road running through a mountain landscape with occasional trees and rocks beside the road (see Fig. 1). We applied detailed texture to the road and surrounding grounds because textured ground is crucial for the perception of optic flow in virtual reality (Cheng et al., 2014) and we displayed a timer above the road, showing the remaining trial time. The timer started at 20:20, with the initial 20 s serving as a period to gain speed before the trial began. We created a Bluetooth coupling between the power output from the bicycle trainer and the velocity of the unfolding of the landscape, creating a sense of power-dependent rectilinear self-motion along the virtual road. We implemented this power output – visual speed coupling to create a realistic virtual experience and to enable a dynamic interplay between optic flow and cycling effort.

During the windowed trials, either the proximal or distal parts of the virtual environment were occluded by a black square. In the distal condition, the remaining visual field included the end of the road and surrounding sky, mountains, rocks, and trees (see Fig. 2a). In the proximal condition, the visual field only contained the road and surroundings directly in front of the cyclist (see Fig. 2b). The timer was positioned to be within the cyclist's view. Every 2 min, an RPE item (Borg CR10; Borg, 1998; Foster et al., 2001) and PM item were displayed in front of the cyclists' view (see Fig. 3). We used the 10-point Borg CR10 as it is a reliable equivalent of the more commonly used 6-to-20 RPE scale (Arney et al., 2019) and maintains consistency in scale range compared to the PM item. We adapted the PM item from an existing measure of PM experiences used in rowing races (Den Hartigh et al., 2016), modifying it to align with the context of our study by rephrasing the objective to focus on cycling power rather than rowing performance. Additionally, we switched from a 1-to-9 scale to a 0-to-10 scale to align with the Borg CR10. After the participants provided their responses, the items disappeared.

Finally, because strong differences in posture between the proximal and distal window conditions might have a relation with perceived effort and bodily responses (on bicycle trainers, more upright postures sometimes have beneficial effects on performance, e.g., Evangelisti et al., 1995; Fintelman et al., 2016; Gnehm et al., 1997; Grappe et al., 1998; Jobson et al., 2008), we measured postural movements throughout the trials using a Vicon motion capture system. The system comprised 8 infrared cameras recording movements at 100 Hz (Vantage V5, lens 8.5 mm, Oxford Metrics, UK). Markers were placed on the neck (second thoracic vertebrae), the back (the midpoint between inferior angles of the most caudal points of the scapulae), and the left and right shoulder (the acromial edges of the scapula). Additional markers were placed on the bicycle stem bolt and the head-mounted display. We used Vicon Nexus 2 software to save the three-dimensional position of each marker over time. The motion capture system recorded the three-dimensional position of the participants' head, neck, shoulders, and back, which we used to calculate the average vertical position of each body part per trial. We used this to explore potential associations between the vertical positions of the head, neck, shoulders, and back, and the average exerted and perceived effort.

2.3. Procedure

The study consisted of three sessions, each lasting approximately 1 h, with a minimum of 3 days between the sessions to allow for sufficient recovery time. In each session, participants performed one of three 20-min cycling trials: first the baseline trial, followed by the distal and proximal trials in counterbalanced order. The participants were instructed to abstain from consuming food or caffeine for 1 h prior to each session. At the beginning of each session, participants completed a questionnaire regarding their sports activities in the past 7 days, as well as their food and alcohol consumption on the day of the session. The purpose of this was to check whether any abnormal activities were reported that could hinder cyclists' participation. Prior to each trial, we informed participants that they could stop the trial at any time if they experienced any cybersickness symptoms. This occurred once, and the participant did not finish the study. Next, the motion capture markers were placed on the participants' bodies, and the head-mounted display was calibrated to align their head position with the head position in VR. The participants then performed a 5-min warm-up of self-paced cycling on the bicycle trainer while being exposed to the complete virtual environment. After the warm-up, we provided them with water and gave them a 3-min break before the trial began. This procedure was similar for each session.

2.3.1. Baseline session

In the baseline session, participants completed a questionnaire about age, gender, weight, height, and cycling habits and

Fig. 1. Experimental Set-up and Virtual Environment.

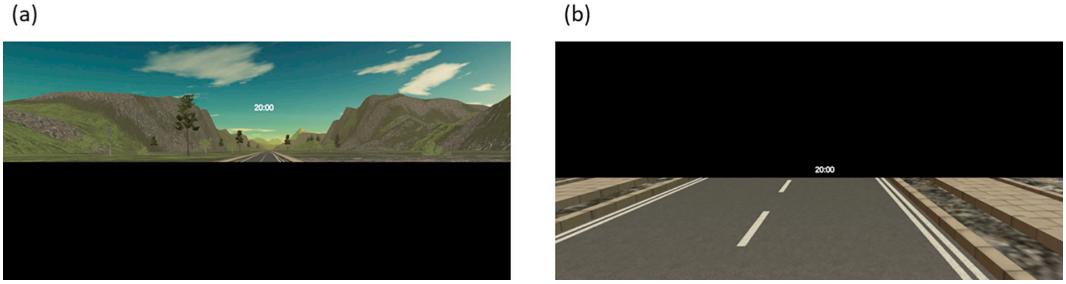

Fig. 2. Virtual Environment in the (a) Distal and (b) Proximal Conditions.

PM. At this moment I have the impression of progressing towards my power objective	
0	Not at all
1	
2	
3	
4	
5	Moderately
6	
7	
8	
9	
10	Very strongly

Borg CR10. Level of effort	
0	Nothing at all
1	Very light
2	Light
3	Moderate
4	Somewhat hard
5	Hard
6	
7	Very hard
8	
9	Extremely hard (almost maximal)
10	Maximal

Fig. 3. PM and Borg CR10 Items Displayed in the Cyclists' View.

experience. We then gave them a tour of the experiment room and explained that the goal of the experiment was to investigate the effect of VR on their performance, sensations, and movements on the bicycle. We explained that they would carry out three cycling trials during which we would measure their power output and cycling posture. We told them that the first trial would take place during this session, and their goal would be to produce the highest possible average power output for 20 min while viewing a virtual environment. These instructions resemble the 20-min functional threshold power (FTP^{20}) test, which is commonly used by cyclists (Mackey & Horner, 2021). We explained that the goal in the following sessions would be similar but slightly different. After these explanations, they performed the warm-up, followed by the break and the baseline trial. During the baseline trial, they were exposed to the complete virtual environment, and after the trial, they were informed about their average power output and were told that the goal in the next two sessions was to surpass their average baseline power output. This approach was designed to create a clear, challenging, individualized goal, that could effectively trigger PM.

2.3.2. *Proximal and distal sessions*

In the second session, we introduced the RPE and PM items. The participants were informed that both these items would appear every 2 min during the trial, after which they would have to verbally provide a number from 0 to 10 in response to the items. We instructed the participants to answer clearly, even when they would become tired. We also assured them that there were no right or wrong answers, and that they could choose a number that best reflected their own feelings. Prior to the trial, the participants familiarized themselves with the items and practiced answering them during the warm-up. We explained that the RPE item measured their overall level of perceived exertion at that moment, and we provided examples of possible scores from Borg (1998). We explained that the goal of the PM item was to assess their momentary perception of progressing toward surpassing their average baseline power.

Additionally, we asked them to minimise postural differences between the two conditions, with the inevitable exception of head movements. We showed the participants the proximal and distal conditions and asked them to position themselves in a way that allowed them to see both windows. Before the trial, we reminded them of their baseline power and emphasised the importance of trying their best to surpass it. After these explanations, the participants performed the trial in either the proximal or distal condition, which were counterbalanced to avoid order effects. The items first appeared at the 1-min mark, and subsequently at intervals of 2 min. The order of appearance of the items was counterbalanced. After responding, the first item disappeared, followed immediately by the appearance of the second item, which also disappeared after responding. We recorded the given responses.

In the third and final session, we reminded the participants of the items and practiced these during the warm-up. Before the trial, we reminded them to position themselves in a way that allowed them to see both the proximal and distal windows and again reminded them of their baseline power, emphasizing the importance of trying their best to surpass it. The participants were kept unaware of their performance in the first windowed trial. After the warm-up and break, the participants performed the remaining windowed trial with the items appearing in the same order as in the first windowed trial. At the end of the session, we asked the participants to identify their perceived best trial and to explain why. Subsequently, they received their results and were debriefed about the experiment.

2.4. Data analysis

As a first check, we assessed any meaningful disparities between the conditions in sports activities, food, and alcohol consumption. To analyse postural differences we conducted one-way repeated measures ANOVAs with condition (proximal vs. distal) as a within-subjects variable and average vertical positions as dependent variables. To investigate the relationship between posture and perceived and exerted effort, we conducted repeated measures correlation analyses using the rmcorrShiny app (Marusich & Bakdash, 2021).

Our main data analysis involved several steps to examine the effects of proximal and distal window conditions on cyclists' exerted and perceived effort and the possible role of PM. For each trial, approximately 2900 data points of power output in Watts were used to calculate an average exerted effort. In the windowed trials, we recorded 10 responses on both the RPE and PM item and calculated the average scores of perceived effort and PM experience for each trial.

We performed a one-way repeated measures ANCOVA to assess the impact of the optic flow conditions on average exerted effort (Hypothesis 1). We included the average power output from the baseline trial as a covariate to the model to control for potential individual differences at the baseline level. In the event of finding a significant interaction effect between optic flow condition and baseline power, we planned to perform a Johnson-Neyman procedure to determine the range of baseline power values within which the effect of optic flow conditions on exerted effort would be statistically significant (Ji, 2016; Montoya, 2019). Subsequently, we performed a one-way repeated measures ANOVA to assess the effect of the optic flow conditions on cyclists' average perceived effort. Lastly, we performed a one-way repeated measures ANOVA to test the effect of the optic flow conditions on PM and a repeated measures correlation analysis to assess the relation between PM and exerted effort (Hypothesis 2).

3. Results

3.1. Preliminary checks

Paired samples *t*-tests revealed no significant differences between the conditions in training hours over the past 7 days, days since the last training, and alcohol consumption on the day before the session ($ps > 0.11$). A McNemar's test showed no significant difference between the conditions' distributions of time since the last meal ($p = .453$). Repeated measures correlation analyses between average vertical body positions and average exerted and perceived effort showed no significant correlations among the vertical positions and exerted and perceived effort ($r_{rm}s < 0.30$, $ps > 0.10$).[1]

3.2. Exerted and perceived effort

A one-way repeated measures ANCOVA, with average baseline power ($M = 203.55$, $SD = 45.89$ W) as a covariate and average exerted effort as the dependent variable, revealed a significant main effect of optic flow condition, $F(1,28) = 8.85$, $p = .006$, $\eta_p^2 = 0.24$. On average, exerted effort was higher in the proximal condition ($M = 206.69$, $SD = 45.89$ W) than in the distal condition ($M = 203.91$, $SD = 49.62$ W). However, the repeated measures ANCOVA also showed a significant interaction effect between optic flow condition and the covariate baseline power, $F(1,28) = 7.23$, $p = .012$, $\eta_p^2 = 0.21$, indicating that the effect of the optic flow conditions on exerted effort depends on participants' baseline power.

To further investigate this interaction, we conducted a Johnson-Neyman procedure to identify the range of baseline power values within which exerted effort in the optic flow conditions significantly differed (Ji, 2016; Montoya, 2019). Fig. 4 visually illustrates the regions of significance as identified by the Johnson-Neyman procedure ($\alpha = 0.05$). As shown, the slope of the conditional effect of optic flow condition becomes non-significant when baseline power exceeds 195.85 W and becomes significant again when it exceeds 340.82 W.[2] Specifically, for baseline power values lower than 195.85 W, the 95 % confidence interval for the slope of the conditional effect (i.e., proximal - distal) only included values above zero, indicating a significant beneficial effect of the proximal condition. Additionally, the analysis revealed that exerted effort was significantly lower in the proximal condition compared to the distal condition when baseline power was above 340.82 W. Specifically, for baseline power higher than 340.82 W, the 95 % confidence interval for the conditional effect only included values below zero, indicating a significant beneficial effect of the distal condition. Thus, our findings suggest that while the proximal condition demonstrates a beneficial effect on exerted effort, this effect disappears as baseline power increases, and even reverses at the highest levels of baseline power.

A one-way repeated measures ANOVA showed no significant effect of optic flow condition on average perceived effort, $F(1,29) = 0.02$, $p = .833$, $\eta_p^2 = 0.00$ (proximal: $M = 6.38$, $SD = 1.22$; distal: $M = 6.39$, $SD = 1.31$).

3.3. Psychological momentum

A one-way repeated measures ANOVA indicated a non-significant difference in PM experience between the proximal condition ($M = 5.29$, $SD = 1.56$) and the distal condition ($M = 4.90$, $SD = 1.88$) in the expected direction, $F(1,29) = 2.07$, $p = .161$, $\eta_p^2 = 0.07$. The

[1] A more detailed overview of one-way repeated measures ANOVAs on postural differences, and repeated measures correlations between vertical body positions and exerted and perceived effort is included in the Supplementary Materials (See S1).

[2] A more detailed overview of the conditional effect at different levels of baseline power is included in the Supplementary Materials (See S2).

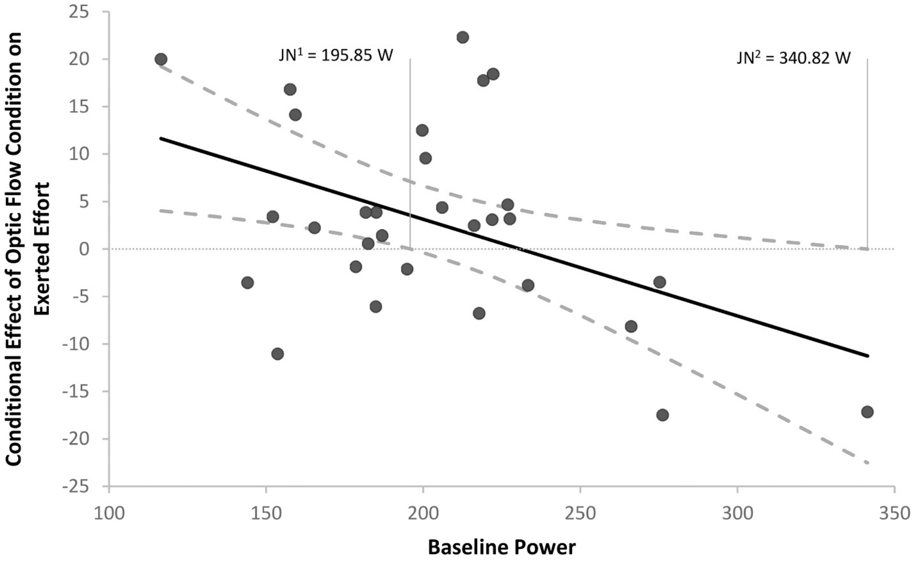

Fig. 4. Conditional Effects of Optic Flow Condition on Exerted Effort (Proximal – Distal) as a Linear Function of Baseline Power. *Note.* The scatter points represent the difference in exerted effort between the proximal and distal conditions across levels of baseline power, for each cyclist. The Johnson–Neyman points (JN) indicate the transition points from significance to non-significance based on an alpha-level of 0.05.

repeated measures correlation between exerted effort and PM was marginal and not significant, $r_{rm} = 0.04$, $p = .814$. These findings suggest no clear evidence of an effect of PM on the relationship between the optic flow conditions and exerted effort. For a visual representation of individual differences in perceived effort and PM across conditions, see Supplementary Materials (Figure S3).

4. Discussion

With the present study we aimed to investigate the effects of exposure to proximal and distal areas of the optic flow field on effort exertion in the context of individual cycling trials. By examining the relationship between optic flow exposure, exerted and perceived effort, and PM, we sought to contribute to the understanding of how different areas in optic flow might differently influence effort regulation during cycling trials. The hypothesis was that, because exposure to more proximal elements in the visual field would result in exposure to faster flow velocities, effort would be judged as stronger and goal progression would be experienced as higher. This sense of progress would contribute to a positive PM experience and increased effort exertion.

4.1. *Exerted effort*

Partially in line with our expectation, we observed that exerted cycling effort was influenced by proximal and distal exposure. However, our findings indicate that the relationship between optic flow exposure and exerted effort is more nuanced than originally hypothesized, and is dependent on cyclists' baseline power. Specifically, we found that proximal exposure significantly increased exerted effort for cyclists with baseline power levels below 195.85 W. However, this significant effect disappeared as baseline power increased, and cyclists with baseline power above 340.82 W even exerted significantly more effort under the distal condition. Notably, only one cyclist in this study exceeded such a high average power output in the baseline, which means that the latter effect should be interpreted with caution.

The beneficial effect of proximal optic flow exposure for relatively lower-level cyclists contrasts with Parry et al.'s (2012) findings, where increased effort exertion was associated with slower optic flow for moderately trained triathletes. This discrepancy may arise from differences in context: while Parry et al. (2012) had participants match baseline efforts, our study asked participants to outperform their baseline effort. From a perception-action perspective, in the context of matching efforts exerted during a baseline trial (Parry et al., 2012), a manipulated slower optic flow likely informs the cyclist to speed up. However, with the aim to outperform the average power output in the baseline trial, proximal exposure tended to increase cycling effort, which came to the fore in the cyclists with relatively lower baseline output. These findings support the idea that an athlete's goals or task demands can alter the relationship between environmental information and effort exertion, as proposed by Smits et al. (2014). More broadly, this aligns with the notion that effort regulation results from a complex interaction between sensory input and task expectations. Specifically, while optic flow may influence effort, it does so within a larger regulatory system that also incorporates psychological factors such as emotions, expectations, and prior experiences, and physiological factors such as afferent feedback from the cardiovascular, respiratory, and musculoskeletal systems, and body temperature (Bruce et al., 2019; Marcora, 2009; Tucker, 2009; Williamson, 2010).

The higher exerted effort during proximal exposure may be due to a faster perception of egospeed. Banton et al. (2005) demonstrated that observers judge their speed higher and more accurately when gazing proximally. The farther the point of focus, the less the perception of speed corresponds to the observer's actual physical speed, whereas the closer the point is to the observer, the more the perceived speed corresponds to the physical speed. In our study, faster flow suggests that cyclists are moving at a higher speed, possibly resulting in a spiralling effect where cyclists increase their effort. Conversely, it could be the case that slower optic flow implies a slower pace, potentially leading to a spiralling effect of decreased effort. In contrast, when the goal is to walk comfortably, the velocity of optic flow generally would lead to adjustments in walking speed in the opposite direction. When optic flow is manipulated to be faster, individuals tend to slow down, while slower manipulated optic flow leads to speeding up, in an effort to compensate for the mismatch between perceived and physical speed (Bardy et al., 1992; Baumberger et al., 2000; Guerin & Bardy, 2008; Mohler et al., 2007; Pailhous et al., 1990).

4.2. Perceived effort and psychological momentum

Contrary to our hypothesis, no significant difference in perceived effort was observed between the proximal and distal conditions. This is not in accordance with Parry et al. (2012), who showed that slower optic flow reduced perceived effort. This discrepancy may be explained by methodological differences. Parry et al. (2012) used 20-km time trials and found that only the slower condition led to differences in perceived effort, as only the lower perception of egospeed and longer perceived time-to-completion posed a threat to task completion. The fast condition, specifying higher egospeed and shorter time-to-completion, did not induce an effect. Similar trends were identified in a study where cyclists' time-to-exhaustion was longer when a manipulated clock ran slower, but unaffected when the clock ran faster (Morton, 2009). In our study, task completion was not threatened as each trial lasted 20 min and the remaining trial time was visible to the cyclists. Consequently, optic flow exposure could not influence cyclists' perception of time-to-completion, plausibly accounting for the absence of significant changes in perceived effort.

Next, we explored the role of PM in the effect of optic flow exposure on exerted effort. While there was a trend in the expected direction, optic flow conditions did not significantly differ in PM experience, nor did PM experience significantly relate to exerted effort. This suggests that the relationship between optic flow exposure and exerted effort is not explained by perceived goal progression. One reason may be that previous studies on PM in cycling focused on inter-individual competitions, which provided clear goal progression relative to an observable opponent (e.g., Briki et al., 2013; Perreault et al., 1998). In contrast, our study focused on visual perception in individual trials, making the sense of goal progression relative to a baseline trial less explicit. Furthermore, PM is a complex process that involves cognitive, affective, motivational, physiological, and behavioural changes (Briki et al., 2013; Den Hartigh et al., 2014; Den Hartigh et al., 2016; Gernigon et al., 2010), making it challenging to capture with a single question. While our study did not show a significant difference in cyclists' experience of goal progression, momentum-like responses could have been present without the cyclists being able to express it. Interestingly, despite the lack of differences in perceived effort, participants with lower baseline power output, in particular, exerted more effort in the proximal condition. For these participants, this could indicate they were experiencing being on a roll, where they could exert more effort without the sensation of higher effort, which corresponds to the experience of positive PM (e.g., Adler, 1981; Briki et al., 2012). Further research with more comprehensive measures of PM could further clarify the interplay between optic flow exposure, PM, and effort exertion in cycling.

4.3. Limitations and future directions

A limitation of this study is that we were unable to precisely quantify the difference in flow velocity between the two conditions. Although flow velocity was certainly higher in the proximal condition, the exact difference would be unique to each cyclist due to the dynamic coupling between power output and optic velocity. Although this dynamic interplay was necessary, it made controlling virtual velocity impossible. This level of detail was not yet the focus of our study, but given the interesting differences between the conditions, it could be an interesting focus for future research. Future research could also integrate eye-tracking technology into the head-mounted display to monitor gaze behaviour throughout the trials, allowing for a more direct examination of the connection between exerted effort and the flow velocity around each point of gaze.

Another limitation concerns the variability in cycling experience among participants, which ranged from regular cyclists to those who cycle only once a week. Future research could benefit from recruiting a more homogeneous group of higher-level cyclists to test the potential beneficial effects of distal optic flow exposure in elite cycling.

Beyond these limitations, there are several interesting directions for future research. One avenue involves implementing VR manipulations that more closely resemble real-world cycling. While occluding proximal and distal areas allowed us to isolate their effects, this type of visual occlusion does not occur in real-world cycling. Future studies might explore alternative methods of manipulating optic flow exposure, such as maintaining the entire virtual environment visible while guiding participants to gaze through proximal or distal frames. This approach would provide a more realistic VR experience, which is known to influence perceptual and affective responses during exercise (Bird et al., 2024).

Other promising future directions include incorporating additional virtual elements such as virtual hills or opponents. Previous VR cycling studies that manipulated virtual hill gradients without altering pedalling resistance found that steeper virtual hills led to higher perceived and exerted effort, likely due to anticipation of increased task demands (Finnegan et al., 2023; Runswick et al., 2023). Cyclists may anticipate greater resistance when encountering a steeper hill, leading to increased effort even though pedalling difficulty remains unchanged. Interestingly, different hill gradients could also be used to manipulate the coupling between exerted effort and optic flow velocity. For instance, a steeper hill would naturally result in slower flow velocity. Beyond terrain manipulations, other VR elements, such as virtual opponents, could also affect effort exertion (Konings & Hettinga, 2018). Perceived speed relative to virtual opponents may influence cyclists' perceptions of goal progress, potentially affecting their exerted effort (cf. Briki et al., 2012; Den

Hartigh et al., 2014).

5. Conclusion

This study provides initial insights into the effects of exposure to proximal and distal areas of the optic flow field on effort exertion in cycling trials. While we found that proximal exposure leads to increased exertion for cyclists with lower average baseline power output, this effect disappeared as the average baseline power increased, and even reversed in the highest region of baseline power output, suggesting that cycling level is an important factor. We did not observe significant differences in perceived effort and PM. This suggests that additional factors, such as a spiralling effect, may influence effort exertion during cycling. Following this first step, we encourage researchers to further explore possible mechanisms of the relation between (proximal and distal) optic flow and effort exertion in the sport and exercise context. Future studies could benefit from monitoring flow velocity, utilizing a complete field of view, and examining gaze behaviour through eye-tracking technology to better understand the role of visual perception in effort regulation.

From a practical perspective, our findings suggest that proximal exposure may benefit amateur cyclists, while distal exposure may be more advantageous for high-level cyclists. Investigating how voluntary or guided gazing toward different areas of the optic flow field influences exerted effort might potentially lead to the identification and training of innovative visual strategies for cyclists and other endurance athletes and may be useful for R&D programs, in which tools can be developed to assist athletes in their gazing (e.g., smart glasses or helmet smart visors). For instance, proximal or distal gazing could be stimulated to enhance effort exertion during training and time trials. While embracing such visual strategies into cyclists' training regimens might result in slight enhancements of effort exertion, they could potentially yield substantial performance improvements. As illustrated by Greg LeMond in 1989 by winning the final time trial and thereby the Tour de France by a mere 8-s margin: In the realm of high-performance sports, even the slightest advantage in effort could be the difference between victory and defeat.

CRediT authorship contribution statement

Sem Otten: Writing – original draft, Visualization, Software, Methodology, Investigation, Formal Analysis, Data Curation, Conceptualization. **Ruud den Hartigh:** Writing – review & editing, Validation, Supervision, Methodology, Conceptualization. **Frank Zaal:** Writing – review & editing, Validation, Supervision, Methodology, Conceptualization. **Benoît Bardy:** Writing – review & editing, Validation, Project Administration, Conceptualization. **Christophe Gernigon:** Writing – review & editing, Validation, Supervision, Project Administration, Methodology, Conceptualization.

Declaration of competing interest

We report there are no competing interests to declare.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.humov.2025.103353.

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